

Benchmark: 2D Hatano-Nelson model

This example is a benchmark of the codes based on the 2D Hatano-Nelson model, where closed-form solutions are available for both amoebic GBZ and strip GBZ (see *arXiv.2506.22743*).

In this example, the following features are illustrated:

- Create a characteristic polynomial by coefficients and degrees.
- Calculate the GBZ subsets with respect to a given eigenenergy E , including the amoebic GBZ and the strip GBZ.
- Data readout for `LineSubset` and `PointSubset`.
- Sweep over an energy regime to get the full GBZ.

Introduction

The Hamiltonian of the 2D Hatano-Nelson model reads,

$$H = \sum_{i,j} \left(J_{x1} c_{i+1,j}^\dagger c_{i,j} + J_{x2} c_{i-1,j}^\dagger c_{i,j} + J_{y1} c_{i,j+1}^\dagger c_{i,j} + J_{y2} c_{i,j-1}^\dagger c_{i,j} \right),$$

where $c_{i,j}$ is the annihilation operator at the site (i, j) , and $J_{x1}, J_{x2}, J_{y1}, J_{y2} \in \mathbb{C}$. In general, the coefficients can be factorized into an ‘‘Hermitian’’ part and a ‘‘non-Hermitian’’ part, i.e.,

$$J_{\alpha 1} = e^{\gamma_\alpha + i\delta_\alpha} J_\alpha, \quad J_{\alpha 2} = e^{-\gamma_\alpha + i\delta_\alpha} J_\alpha^*, \quad \alpha = x, y$$

where $\gamma_\alpha, \delta_\alpha \in \mathbb{R}$ are the non-Hermitian factors, and $J_\alpha \in \mathbb{C}$ is the Hermitian factor.

In this model, the amoebic GBZ and the strip GBZs along x, y and $[11]$ axes have closed forms [1]. The x -strip, y -strip and amoebic GBZs are identical, which read,

$$\beta_x = \exp(\gamma_x + i\theta_x),$$

$$\beta_y = \exp(\gamma_y + i\theta_y),$$

as θ_x, θ_y traverses $[0, 2\pi]$. However, the $[11]$ -strip GBZ is different from the above, which reads,

$$\tilde{\beta}_{[11]} = e^{\gamma_x + \gamma_y + i\theta_{[11]}},$$

$$\tilde{\beta}_y = e^{\gamma_y + i\theta_y} \sqrt{\frac{J_x^* e^{i\Delta_{xy} + i\theta_{[11]}} + J_y}{J_x e^{i\Delta_{xy} - i\theta_{[11]}} + J_y^*}}.$$

Bibliography

- [1] C. Wang, J. Pi, Q. Liu, Y. Li, and Y.-C. Liu, General Theory for Geometry-Dependent Non-Hermitian Bands, Arxiv (2025).